

ABHRADEEP DAS

+91-9007957999 | abhradeep.das@gwu.edu | LinkedIn | GitHub

SUMMARY

Analytics Engineer with 1+ years of experience building tested, documented data models in dbt and SQL on warehouses like Snowflake, and delivering them as governed, self-service Power BI and Tableau dashboards. Proven track record turning multi-source data into a trusted single source of truth at The World Bank and GWU.

EDUCATION

The George Washington University <i>Master of Science, Data Science</i>	May 2025 Washington DC
Techno India University <i>Bachelor of Technology, Computer Science & Engineering</i>	Jul 2023 Kolkata, India

WORK EXPERIENCE

Data Engineer | Saayam for All | Remote Jun 2025 – Present

- Designed and orchestrated modular **Python**, **DBT** and **SQL** transformation models on **Snowflake** using **Apache Airflow**, consolidating multi-source donor and volunteer data into a trusted, centralized data model that served as the single source of truth for downstream reporting.
- Improved donor data reliability by 18% by developing an automated data quality framework in **DBT** and **SQL** with completeness, consistency, and null-check tests, and building SLA-tracking scripts to triage model failures and reduce mean time to detection for data incidents.
- Reduced manual reporting effort by 25% across 5 regions by building a **star schema** (fact tables for donations/activities, dimension tables for donors/locations) and **Power BI** semantic models with **DAX** measures, partnering with cross-functional stakeholders to align metric definitions and enable self-service analytics.

Data Engineer Fellow | The World Bank | Washington, DC Jan 2025 – May 2025

- Improved investment allocation efficiency by 14% by building a climate-prioritization model with Factor Analysis (dimension reduction), Latent Class Analysis (segmentation), and Item Response Theory (measurement/scaling) to rank the highest-return climate-governance reforms across 182 countries, visualized in **Tableau**.
- Engineered **Python**-based **ELT** with **Selenium** to extract web pages, normalize, and quality-validate semi-structured data (JSON, CSV, PDF) from 10+ web sources into standardized, structured inputs for downstream modeling.
- Eliminated manual policy-document review and reduced reliance on external LLM tools by developing an in-house **ML classification** system with **Python** and **Llama 70B** to tag 500+ policy documents by development theme, modeling the output on **Snowflake** and combining it with the earlier indicator analysis in a **Tableau** dashboard that surfaced insights to analysts 20% faster.

Research Assistant I | The George Washington University | Washington, DC Feb 2024 – May 2024

- Automated an end-to-end risk-analysis pipeline to ingest and parse ~1 TB of 10-K SEC filings from **AWS S3**, replacing manual processing and enabling scalable, repeatable financial risk analysis.
- Built an **ETL** workflow in **Python (Regex, spaCy)** to extract Risk Factor 1A passages from raw filings, transforming unstructured text into structured, query-ready tables persisted to S3 for scalable downstream access.
- Integrated a transformer-based text-similarity module into the pipeline to flag anomalous risk-disclosure shifts across companies and quarters, reducing manual review effort and accelerating investment-risk assessments for downstream teams.

Research Assistant I | Calcutta University | Kolkata, India Aug 2022 – Dec 2022

- Collaborated with a cross-functional team using **Git** and **JIRA** to design and deploy a **Docker**-containerized machine-learning system on **AWS** for real-time phishing and fraud detection in text messages and emails, which enabled immediate identification of threats and improved response speed.
- Processed 10,000+ messages monthly using advanced classification algorithms (**XGBoost, SVC**) and NLP techniques with **Sentence-BERT** embeddings, achieving 35% reduction in fraud cases.
- Architected scalable data pipeline using **Python, BeautifulSoup, and Regex** for web scraping and email text extraction, developing NLP classification model with 85% F1 score in fraud detection.

PROJECT EXPERIENCE

Quora Question Pair Similarity | Academic Project Sep 2024 – Dec 2024

- Developed a scalable **ML** solution for duplicate question detection, achieving 93% accuracy through an ensemble approach combining **BERT** embeddings, **XGBoost**, and **TF-IDF**.
- Reduced false positives by 40% using contrastive learning and Siamese networks, enhancing Quora's user experience and search relevance.
- Deployed on **AWS Lambda** with **ONNX** optimizations, decreasing inference latency significantly and enabling scalable real-time predictions

TECHNICAL SKILLS

Programming Languages & Databases: Python, R, SQL(MySQL, PostgreSQL), NoSQL(MongoDB).
Data Transformation & Modeling: dbt(core, tests & docs), Jinja, MetricFlow, Star/Snowflake Schema, SCD Type 2, Elementary
Orchestration & DevOps: dlt, Airflow, Dagster, DBT scheduler, Git / GitHub Actions (CI/CD), JIRA, Docker, Kubernetes
Warehouses & Cloud: Snowflake, BigQuery, DuckDB, AWS, GCP
Business Intelligence & Reporting: MS Excel, Power BI (DAX, Power Query), Tableau, Looker Studio
Data Preprocessing & Analytics: Numpy, Pandas, Matplotlib, Seaborn, Plotly, SpaCy, BeautifulSoup, Selenium
ML/DL Libraries & Frameworks: PyTorch, TensorFlow, Scikit-Learn, Keras, Hugging Face, LLM Integrations (Llama 70B)
Soft Skills: Effective Communication, Analytical Problem Solving, Team Collaboration, Financial Modeling, Time Management